

Methods: detection of coordinated calcium events

Input. A *coordinated event* is a group of calcium-event onsets in several regions of interest (ROIs, one cell each) within seconds of one another; a detector reports each as a *call*, an interval of time. The input is the imaging pipeline’s per-ROI event table: each event’s onset, taken as its half-rise time (t_{50}), on a 0.1 s frame grid. The pipeline exports two event sets, *fast* and *slow*, assigned upstream; a third, *combined*, stream is the time-sorted union of each ROI’s fast and slow onsets, near-coincident pairs kept. *Windows* are each recording’s baseline and treatment periods. The data are 66 recordings (2,109 ROIs, 36 mice) whose first treatment was tetrodotoxin (TTX) or senktide (a neurokinin-3 receptor agonist), in four groups, counted in recordings: diestrus females (DI, 17), ovariectomized females (OVX, 17), males (MALE, 13) and orchidectomized males (ORX, 19). Group is nested in imaging day.

Event floor. A coordinated event needs at least the *event floor* of ROIs co-active within 2 s: the smallest count chance reaches at most once per hour, and never fewer than 3 ROIs. Chance comes from 1,000 draws that shift each ROI’s onset train as a whole by its own offset, uniform in $[-20\text{ s}, 20\text{ s}]$, keeping each ROI’s rate and intervals (after Pipa et al., 2007, 2008, with a larger shift, and edges dropped rather than wrapped). The null keeps coordinated onsets, so coordination-rich windows get a higher, conservative floor. Each window of each stream, and each simulated recording, has its own floor; treatment windows are also scored under their baseline floor. The floor is CoactDetect’s minimum count, and planted events under it are left out of scoring.

Parameters for simulation. Detectors were tuned and scored on simulated recordings with known (*planted*) coordinated events, measured on baseline windows, groups pooled. *Spacing:* coordinated events were located, without the detectors compared here, as runs of the co-active count at or above the floor (peaks under 2 s apart merged). Median gaps between neighbors were 41.3 s (fast), 25.1 s (slow) and 24.8 s (combined), at 9.7, 22.0 and 25.3 events per hour of baseline (gaps are skewed). *Participation:* median participants per moment with at least 4 ROIs co-active within 1 s, over median ROI count (fast 0.20, slow 0.38, combined 0.25), set the middle of three planted levels; the outer two were chosen, and the lowest falls under the floor, so it is not scored. *Background:* quiet and busy per-ROI event rates are the 25th and 75th percentiles across recordings, coordinated share subtracted, varied across ROIs and over time by Gamma factors fitted to the data. *Timing spread:* the standard deviation (SD) of onsets within an event (fast 0.105 s, slow 0.131 s, combined 0.150 s) came from the half-width of the cross-ROI onset correlogram (Perkel et al., 1967), calibrated on simulations of known spread.

Simulated recordings. Each has 32 ROIs and lasts 45 min. It plants the stream’s measured event rate over 45 min (fast 7 events, slow 17, combined 19) over the three levels, laid end to end at gaps resampled from the pooled gaps and placed at a random time, plus six *decoys* (bursts built like middle-level events, not planted events), on a quiet or busy background. Each seed also yields a *no-coordination recording* (quiet, nothing planted) and an *elevated-rate recording* (a 300 s stretch at the 99th percentile of background rate). A variant spaced like ORX checks the rankings. Separate seeds serve *selection*, *held-out* confirmation and *final* scoring, doubled on fast.

Scoring. A call matches a planted event lying in its span or within 2.5 s of it, one to one. A planted event under its recording’s floor, and any call matched to it, are excluded. F1, the harmonic mean of precision and recall, is pooled per background and averaged over the two. Calls on decoys count against precision, and F1 without them is also reported. Calls holding two or more scored events are counted as merges.

CoactDetect. $S(t)$ counts the ROIs with an onset in $(t - w, t]$, w being the count interval. Its mean and variance under a null that circularly shifts each ROI’s onsets within a surrounding *context*, optionally excluding a guard band around t , are computed exactly; t is called when S reaches the floor and $z = (S - \text{mean})/\text{SD}$ exceeds a threshold, set as a normal-tail level α . Calls closer than a merge gap are joined. The design resembles cell-averaging constant-false-alarm-rate radar detection (Finn & Johnson, 1968) and tests of excess coincidence against a locally resampled null (Amarasingham et al., 2012), reached independently. Settings were searched in this sliding form (fast ships fixed 2 s bins) by coordinate search and paired grids from the *shipped* (current) settings: α down to 6×10^{-16} ($z \approx 8$), contexts 20–120 s, guard at most a quarter of the context. The floor is always computed at 2 s, so its once-per-hour guarantee holds only at $w = 2\text{ s}$. Every setting must meet budgets set with headroom over the shipped settings’ measured rates: calls per hour on the no-coordination recording (fast 7, slow 1, combined 10), 1 call per minute in the elevated stretch, and a quiet-to-busy precision difference of at most 0.10. A candidate was adoptable if its held-out F1 gain over the shipped setting had a 95% bootstrap interval above zero, its best values lay inside the tested range, and it met every budget; a best value at a cap, or one that switches a setting off, was reported, not adopted.

Chorus. Chorus, a learned detector, reads a binary ROI $\times$ frame onset raster. A four-layer dilated convolutional stack (Yu & Koltun, 2016) shared by all ROIs filters each ROI; each channel is standardized over the whole input and passed through a sigmoid as a per-ROI vote. Votes are pooled across ROIs (mean, SD, mean of the top four), so any number of ROIs in any order is accepted (Qi et al., 2017; Zaheer et al., 2017), and a dilated head scores each frame; frames over a threshold chosen on training-side validation recordings become calls, runs within 2 s merged. Standardization spans 409.6 s excerpts in training and the whole window at inference. A participation variant sums a separate bounded per-ROI vote into a count, takes the floor as input and learns which ROIs took part. Training was per stream (Adam, Kingma & Ba, 2015; learning rate 0.01; 900 steps; class-weighted binary cross-entropy), with extra events planted at the floor and one ROI either side, sub-floor events labelled negative. Of five initializations, the best on selection recordings within CoactDetect’s no-coordination budget was kept (the best overall if none met it).

Benchmark. Each detector was compared with CoactDetect on the final seeds as a paired per-seed F1 difference with a 95% bootstrap interval over seeds (2,000 resamples; Efron, 1979).

Output. Each call, whatever its detector, is measured alike: onsets within $\pm 1\text{ s}$ (fast) or $\pm 5\text{ s}$ (slow, combined) of the call’s center, widened to its span, are split where consecutive onsets are over 0.5 s (fast) or 2.5 s (slow, combined) apart; the group with most distinct ROIs is the event’s *core*. These lengths were chosen, not measured. The event time is the call’s onset, with the core’s first onset reported beside it (which should be the event time is undecided); *width* is the core’s first-to-last onset span; *amplitude* is core ROIs divided by width floored at one frame (ROIs s^{-1}), undefined for one ROI. Amplitude measures packing, not fluorescence, is dominated by width, and has a provisional name. On simulated data the core’s ROI count never under-counts planted participants and sometimes over-counts, most on combined (median +2–3 ROIs); width and amplitude are unchecked.

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